

Gary D Alvarado

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EDUCATION

University of California, Davis

M.S. in Computer Science

Davis, California

September 2026 - May 2028

California State University, East Bay

Upper-Division Computer Science Coursework

Hayward, California

May 2025 - May 2026

- **Coursework completed:** Software Engineering, Operating Systems, Computer Networks, Analysis of Algorithms, Artificial Intelligence, Computer Architecture, Programming Languages Concepts, Data Structures & Algorithms

California State University of East Bay

B.S. in Business Administration

Hayward, California

Graduated, Fall 2023

- **Relevant Coursework:** IT Management, HR Analytics, Strategic Business Management, Consumer Behavior, Decision Science, Marketing Research, Employee Relations & Law, Operations Management

EXPERIENCE

Hayward Unified School District

Hayward, California

Long Term AP Computer Science Principles/Computer Programming Substitute Teacher

September 2024 – July 2025

- Taught 130+ students across 5 sections (AP Computer Science Principles and Computer Programming), covering fundamental programming concepts using C++, JavaScript, and HTML/CSS, along with core computer science concepts including networking and data representation.
- Designed and led hands-on projects and lessons to prepare students for the AP CS exam and build foundational problem-solving and software development skills.
- Introduced tools such as Github, AI tools, and different programming languages.

PROJECTS

Communication System – Java, Multithreading, Networking, OOP | May 2026

- Collaborated with a team to build a Java client-server communication system supporting multiple concurrent users.
- Created UML diagrams and a Software Requirements Specification (SRS) to guide system design and development.
- Implemented multithreaded server logic to manage concurrent client connections and real-time communication.
- Applied object-oriented design, modular programming, and Git version control throughout the project lifecycle.

Quizzical – React, JavaScript, API Integration | January 2026

- Built an interactive quiz application using React, leveraging state management and hooks for dynamic UI updates.
- Integrated an external trivia API to fetch and display randomized questions and track user responses.
- Implemented conditional rendering and user interaction logic to evaluate answers and display scores.

OCaml Stack-Based Interpreter – OCaml | August 2025

- Built a functional interpreter to execute stack-based commands with parsing for operations like Push, Pop, and Arithmetic Operators.
- Utilized recursion, pattern matching, and immutable data structures for execution logic.

AP Computer Science Student Portfolio Showcase – JavaScript / HTML / CSS | May 2025

- Developed a portfolio platform showcasing 70+ student-built JavaScript projects, linking to GitHub repositories and live deployments.

BFS Flight Path Finder – C++ | April 2025

- Developed a program that reads a .txt file of city-to-city flights and uses Breadth-First Search (BFS) to determine if a path exists between two locations.
- Demonstrated proficiency in file I/O, graph data structures, and queue-based traversal algorithms.
- Emphasized practical application of BFS in solving real-world connectivity problems.

Personalized Dashboard – JavaScript / Chrome Extension / API Integration | January 2025

- Built a Chrome Extension that renders a real-time dashboard on each new tab.
- Integrated multiple APIs to display Bitcoin prices, local weather (via geolocation), time, and dynamic backgrounds.

SKILLS

Programming: C++, Java, JUnit, JavaScript, React, OCaml, HTML, CSS, Python, SQL, Lisp, Object Oriented Programming

Tools: Github, Generative AI, REST API's, Netlify, Linux, Microsoft Office, Google Classroom